(1)

```
BEGIN

  CLASS Person
      Name
      Age
  END CLASS
  CLASS Student INHERITS Person
      RegisterNumber
      Marks[5]

    METHOD CalculateAverage()
        Sum ← 0
        For i ← 1 TO 5
            Sum ← Sum + Marks[i]
          END FOR
        Average ← Sum/5
      END METHOD
    END CLASS
  CLASS ResearchStudent INHERITS students
      ResearchStudent
      method DisplayCompleteDetails()
          PRINT Name, Age, RegisterNumber, Average, Research
        END METHOD
    END CLASS

  CREATE RS AS Researchstudent
  Input all details
  CALL RS.CalculateAverage()
  CALL RS.Display completeDetails()

END
```

② BEGIN

```
READ N
  READ ARRAY A
  For i←0 TO N-1
    For j←i+1 TO N-1
      IF A[i] == A[j]
          DELETE A[j]
      ENDIF
    ENDFOR
ENDFOR
  PRINT modified Array
END
```

③

incorrect line : RETURN length * length

④

Output : -3

⑤

```
BEGIN
READ N
READ A [0....N-1]
first ← -∞
Second ← -∞
 FOR i← 0 TO N-1
     IF A[i] > first THEN
```

```
        second ← first
        first ← A[i]
    ELSE IF A[i] > second AND A[i] ≠ first THEN
        second ← A[i]
    ENDIF
    ENDFOR
  PRINT second
END
```

6

```
BEGIN
INTERFACE payment
    METHOD Pay(amount)
END INTERFACE
CLASS Credicard IMPLEMENTS payment
    METHOD pay(amount)
        PRINT "Payment using credit card"
    END METHOD
END CLASS
CLASS UPI IMPLEMENTS Payment
    METHOD pay(amount)
        PRINT "Payment using UPI"
    END METHOD
END CLASS
```

```
CLASS NetBanking IMPLEMENTS Payment
    METHOD Pay (amount)
        PRINT "Payment using Net Banking"
    END METHOD
END CLASS

READ choice
READ amount
IF choice = 1 THEN
    CREATE Creditcard P
ELSE IF choice = 2 THEN
    CREATE UPI P
ELSE
    CREATE NetBanking P
ENDIF
    CALL P. pay (amount)
END
```

(7)

CLASS DOG . INHERITS Animal

Animal obj ← NEW DOG

(8)

```
CONSTRUCTOR (empId, empName)
    id ← empId
    name ← empName
```

(9)

```
BEGIN
  CLASS vehicle
    DATA
         vehicleNO
         owerName
    METHOD display()
         PRINT vehicleNO
         PRINT overName
     END METHOD
  END CLASS
  CLASS Car INHERITS vehicle
     DATA
         fuelType
     METHOD display()
          CALL vehicle.display()
          PRINT fuelType
      END METHOD
     END CLASS
  CLASS Truck INHERITS vehicle
     DATA
          loadCapacity.
     METHOD display()
          CALL vehicle.display()
           PRINT loadCapacity.
      END METHOD
  END CLASS
```

```
READ N
  FOR i ← 1 TO N
     READ type
  IF type = "Car" THEN
      CREATE Car V
     READ V.vehicleNO
     READ V.ownerName
     READ V.fuelType
  ELSE
      CREATE Truck V
        READ V.vehicleNO
         READ V.ownerName
         READ V.loadCapacity
     ENDIF
        STORE V
   ENDFOR
        FOR EACH V IN vehicleList
            CALL V.display()
        ENDFOR
     END
```